

Rodrigo Alvarez Lucendo

AI Engineer - EPFL & TU Delft

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PROFESSIONAL SUMMARY

AI engineer passionate about turning cutting-edge research into reliable systems, trained at TU Delft and EPFL and experienced at Amazon and an early-stage aerospace AI startup. Second author of an ICML 2026 paper on Transformer learning dynamics, presented at one of the world’s leading AI conferences.

WORK EXPERIENCE

Jun 2025–Present	AI & Machine Learning Engineer , Xcert.ai - First employee at a CHF 1.5M+ pre-seed startup building AI assistants for aerospace certification. - Led the development, monitoring, and evaluation of an agentic workflow using the OpenAI Agent SDK. - Co-created the first aerospace LLM benchmark to evaluate domain-specific model performance. - Led technical customer communication with the European Space Agency (ESA).
Feb–Jul 2025	Selected AI Projects , EPFL AI Tutor: Fine-tuned a 0.6B LLM with SFT and DPO to create an AI STEM tutor. nanoGPT: Built a decoder-only Transformer and BPE tokenizer from scratch, then extended it with MoE. RAG QA: Analyzed how contextual relevance influences question-answering performance. FETCH, EPFL AI Team: Co-developed an LLM-based mission planner for a Unitree Go2 robot using live sensor input, Nav2, and MCP servers.
Jul 2024–Jan 2025	Data Engineer , Amazon (Internship) - Deployed a monitoring tool for machine learning models in production. - Pioneered the team’s exploration of AI agents by building a ReAct agent for natural-language metric analysis. - Delivered complex SQL queries and business analysis for BI teams.
2022–2024	Teaching Assistant , TU Delft & EPFL Guided 300+ students in BSc/MSc courses (ML, Big Data, Deep Learning); Head TA for Computational Intelligence.

EDUCATION

2023–May 2026	MSc Computer Science (AI & ML focus) , TU Delft & EPFL Coursework: Modern NLP, Deep Learning, Graph ML, Deep RL, Computer Vision, Distributed ML Systems.
2019–2023	BSc Computer Science , TU Delft Thesis: Deep learning model for algal bloom forecasting using spatiotemporal data on HPC.

RESEARCH

Feb–Sep 2025	Research Assistant , Theory of Machine Learning Lab, EPFL Investigated how Transformers learn high-order Markov chains through staged dynamics; work accepted at the ICML 2026 Main Conference.
2025–2026	Master Thesis , Natural Language Processing Lab, EPFL Proposed a framework for understanding why Transformer-based models preferentially memorize certain data types; preparing submissions to the NeurIPS 2026 Workshop and ICLR 2027 Main Conference.

PUBLICATION

Incremental Learning of Sparse Attention Patterns in Transformers. Oğuz Kaan Yüksel, **Rodrigo Alvarez Lucendo**, Nicolas Flammarion. Second author; presented at the ICML 2026 Main Conference (regular acceptance). Earlier version at EurIPS 2025 Workshop on Principles of Generative Modeling (PriGM). arXiv:2602.19143.

SKILLS

Languages	Python, Java, SQL, Scala, Bash, TypeScript, English, Spanish.
ML / Deep Tech	PyTorch, Hugging Face, OpenAI Agent SDK, PyTorch Lightning, Weights & Biases.
Systems / Tools	Docker, Kubernetes, AWS, Neo4j, Postgres, FastAPI, Git, HPC.
Interests	Rowing, alpine skiing, strength training.